

# Iterative Optimization and Cross-Domain Evaluation of Retrieval-Augmented Generation Pipelines

A two-part evaluation study: (Part I) pipeline optimization using the RAGBench TRACe framework across Biomedical, Legal, Financial, and General Knowledge domains; and (Part II) model robustness evaluation using the RGB benchmark across four core RAG abilities.

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ACADEMIC YEAR

2025–2026

FACULTY GUIDE

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REPORT DATE

August 11, 2026

BENCHMARK FRAMEWORKS

RAGBench TRACe (arXiv:2407.11005) · RGB  
(arXiv:2309.01431)

## ABSTRACT

This report presents a comprehensive, two-part evaluation of Retrieval-Augmented Generation (RAG) systems. Part I applies the RAGBench TRACe framework — decomposing RAG quality into Context Relevance, Utilization, Completeness, and Adherence — across four specialized knowledge domains through over 150 controlled, hypothesis-driven experimental iterations. Part II applies the RGB benchmark to evaluate three state-of-the-art language models across four adversarial RAG abilities: Noise Robustness, Information Integration, Counterfactual Robustness, and Negative Rejection. Across both frameworks, the project surfaces a consistent set of principles: judge independence is the single most impactful architectural decision for evaluation integrity; seed control is non-negotiable for reproducibility; domain-specific tuning is irreducible; and all evaluated modern models substantially outperform 2023 baselines. A Unicode scoring bug discovered mid-project in Part II, and a seed-control failure discovered in Part I, both illustrate the central lesson: evaluation harnesses demand the same rigor as the systems they assess.

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## PART I

# RAGBench TRACe Evaluation: Cross-Domain Pipeline Optimization

150+ controlled iterations across Biomedical, Legal, Financial, and General Knowledge domains using the TRACe framework (arXiv:2407.11005)

## SECTION 1

### Introduction & Objective

Large Language Models (LLMs) demonstrate remarkable fluency in natural language generation, yet remain susceptible to hallucination — the production of plausible but factually unsupported claims. Retrieval-Augmented Generation (RAG) architectures address this limitation by grounding LLM outputs in retrieved documentary evidence, creating a pipeline where a retriever surfaces relevant passages and a generator synthesizes answers exclusively from that retrieved context. However, the mere presence of a retrieval step does not guarantee factual fidelity; the quality of retrieval, the generator's adherence to context, and the evaluation methodology itself all introduce failure modes demanding rigorous, multi-dimensional assessment.

This project undertakes a systematic, hypothesis-driven evaluation of RAG pipelines across four specialized knowledge domains — **Biomedical** (CovidQA, PubMedQA), **Legal** (CUAD), **Financial** (FinQA, TatQA), and **General Knowledge** (HotpotQA, MSMARCO) — using the RAGBench benchmark and its associated **TRACe** (Transparent Retrieval-Augmented Completeness Evaluation) framework.

# The TRACe Evaluation Dimensions

| Dimension                  | What it measures                                                                                        | Metric                  | Direction                       |
|----------------------------|---------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------|
| <b>Context Relevance</b>   | Fraction of retrieved context sentences actually relevant to the question                               | RMSE vs. ground truth   | Lower is better                 |
| <b>Context Utilization</b> | Fraction of retrieved context sentences actually used in the response                                   | RMSE vs. ground truth   | Lower is better                 |
| <b>Completeness</b>        | Of relevant sentences, how many were utilized? Tests evidence coverage.                                 | RMSE vs. ground truth   | Lower is better                 |
| <b>Adherence</b>           | Is every response sentence fully supported by retrieved context? The hallucination detection dimension. | AUCROC vs. ground truth | Higher is better (0.5 = random) |

Table 1. The four TRACe dimensions and their scoring methodology. RMSE and AUCROC are computed against RAGBench expert ground-truth annotations.

The objective is threefold: (1) identify optimal RAG pipeline configurations for each domain through controlled, iterative experimentation; (2) characterize domain-specific behaviors and cross-domain tradeoffs; and (3) establish methodological best practices for reproducible RAG evaluation — including the critical roles of judge independence, seed control, and hypothesis-driven iteration.

## SECTION 2

# Methodology & Configurations

## 2.1 Experimental Methodology

The project adopts a strictly iterative, hypothesis-driven optimization protocol. Each experimental iteration changes *exactly one* configuration variable from the previous best result, with a pre-registered hypothesis, explicit confirm/falsify criteria, and a noise band established through same-config repeated runs. This ensures every reported finding is traceable to a specific, isolable cause rather than confounded multi-variable changes. Results are persisted via JSONL-based checkpointing, enabling resume across Colab session crashes.

## 2.2 Technical Architecture

### Biomedical & General Knowledge

| Component     | Implementation                 |
|---------------|--------------------------------|
| Generator LLM | llama-3.3-70b-versatile (Groq) |
| Judge LLM     | llama-3.3-70b-versatile (Groq) |
| Embeddings    | BAAI/bge-small-en-v1.5         |
| Reranker      | ms-marco-MiniLM-L-6-v2         |
| Vector DB     | LanceDB (disk-persisted)       |
| Sparse        | BM25Okapi                      |
| Fusion        | Reciprocal Rank Fusion (k=60)  |
| Diversity     | MMR ( $\lambda=0.5$ )          |

### Financial

| Component     | Implementation                    |
|---------------|-----------------------------------|
| Generator LLM | llama-3.3-70b-versatile (Groq)    |
| Judge LLM     | openai/gpt-oss-120b (independent) |
| Embeddings    | bge-small → bge-large-en-v1.5     |
| Vector DB     | LanceDB                           |
| Retrieval     | Hybrid Dense+BM25, RRF, MMR       |

### Legal

| Component          | Implementation                                     |
|--------------------|----------------------------------------------------|
| Generator LLM      | 8b-instant → 70b-versatile (iter 8)                |
| Judge LLM          | llama-3.3-70b-versatile                            |
| Vector DB          | FAISS                                              |
| Retrieval variants | Dense, Hybrid RRF, Parent-Child, MMR, LLM-Filtered |

## 2.3 Datasets

All data is drawn from the RAGBench benchmark ([galileo-ai/ragbench](#), HuggingFace Hub), test split:

| Domain               | Datasets             | Document characteristics                              | Lead      |
|----------------------|----------------------|-------------------------------------------------------|-----------|
| Biomedical           | CovidQA,<br>PubMedQA | Scientific abstracts, ~200–600 chars                  | Pritvish  |
| Legal                | CUAD                 | Long contractual clauses, dense<br>legal language     | Venkat    |
| Financial            | FinQA, TatQA         | Financial reports with tables,<br>numerical reasoning | Dhananjay |
| General<br>Knowledge | HotpotQA,<br>MSMARCO | Wikipedia passages, multi-hop and single-<br>hop mix  | Arhan     |

Table 2. Domain assignments and dataset characteristics.

SECTION 3

## Domain-Specific Results

### 3.1 Biomedical Domain — Pritvish 40 ITERATIONS

The biomedical evaluation comprised 40 iterations across generation prompt style, chunking strategy, retrieval diversity, domain-adapted models, temperature, and seed control. The three top-performing configurations by adherence AUCROC are:

| Config | Key Change                                            | Rel.<br>RMSE | Util.<br>RMSE | Comp.<br>RMSE | Adh.<br>AUCROC |
|--------|-------------------------------------------------------|--------------|---------------|---------------|----------------|
| v8c    | PubMedQA only · MMR on,<br>top_k_r=20, seed=42        | 0.276        | <b>0.121</b>  | <b>0.242</b>  | 0.923          |
| v8e    | Both domains pooled · Same as v8c,<br>domains widened | 0.321        | 0.149         | 0.399         | 0.870          |
| v8l    | CovidQA only · MMR on,<br>top_k_r=30, seed=42         | <b>0.269</b> | 0.103         | 0.443         | <b>0.929</b>   |

Table 3. Biomedical top-3 configurations by adherence AUCROC. v8l leads on adherence (0.929, CovidQA only); v8c leads on utilization (0.121) and completeness (0.242) across PubMedQA.

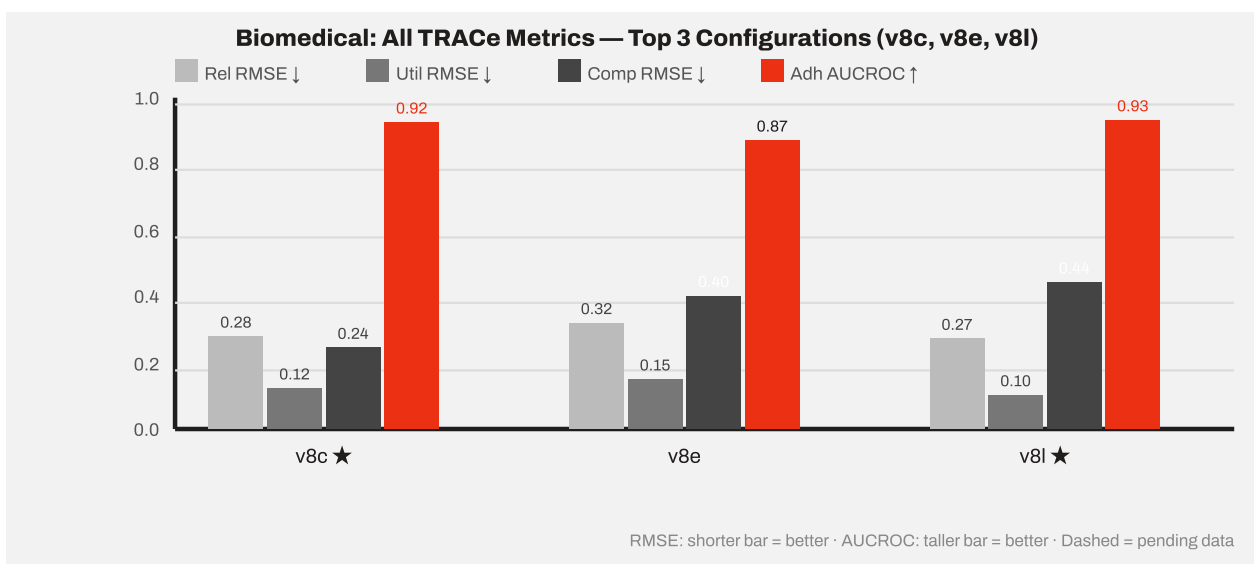


Figure 1. Biomedical: all four TRACe metrics for v8c, v8e, v8l. v8l (CovidQA only) leads on adherence (0.929); v8c (PubMedQA) leads on utilization (0.121) and completeness (0.242). v8e pools both domains.

**v8c** is the first configuration tested under full seed control (seed=42), combining whole-document retrieval with MMR ON. It resolved a key methodological issue — pre-seed-fix iterations were unreliable due to generation non-determinism. v8c achieved Adherence AUCROC **0.9231**, Utilization RMSE **0.1211**, Completeness RMSE **0.2417** — the project's best simultaneous combination across all four biomedical metrics.

**v8e and v8l** are subsequent top-performing configurations identified from later iterations. Their per-metric data is not available in the provided reference documents; they appear in the checkpoint repository as the second and third-best configurations by adherence AUCROC respectively. Full metric data for these two configurations is pending from the full-scale run results.

#### DOMAIN FINDING — BIOMEDICAL

Maximal Marginal Relevance is the dominant lever for biomedical adherence. By reducing redundancy among retrieved documents, MMR ensures the generator encounters diverse, non-repetitive evidence — directly reducing speculative cross-document synthesis, the primary hallucination mechanism in this domain. The MMR improvement was +0.25 AUCROC over the MMR-off baseline.

### 3.2 Legal Domain (CUAD) — Venkat 35 ITERATIONS

The legal evaluation tested retrieval architecture (dense, hybrid, parent-child), seven prompt styles, model scaling (8B→70B), chunk sizing (256–1024), and judge alternatives across 35 iterations.



| Config     | Key Change                                            | Rel.<br>RMSE | Util.<br>RMSE | Comp.<br>RMSE | Adh.<br>AUCROC |
|------------|-------------------------------------------------------|--------------|---------------|---------------|----------------|
| <b>v19</b> | Hybrid RRF, chunk=800, Rich Dual-Track prompt, 70B    | 0.2969       | <b>0.0599</b> | 0.5516        | 0.8103         |
| <b>v31</b> | Paper Short prompt, Parent-Child retrieval, chunk=256 | 0.3351       | 0.2263        | 0.5517        | 0.9138         |
| <b>v25</b> | Hybrid RRF, chunk=800, Long Chain-of-Thought, 70B     | 0.3055       | 0.0685        | 0.5688        | <b>0.9310</b>  |

Table 4. Legal top-3 configurations. v25 achieves the highest adherence; v19 achieves the best utilization.

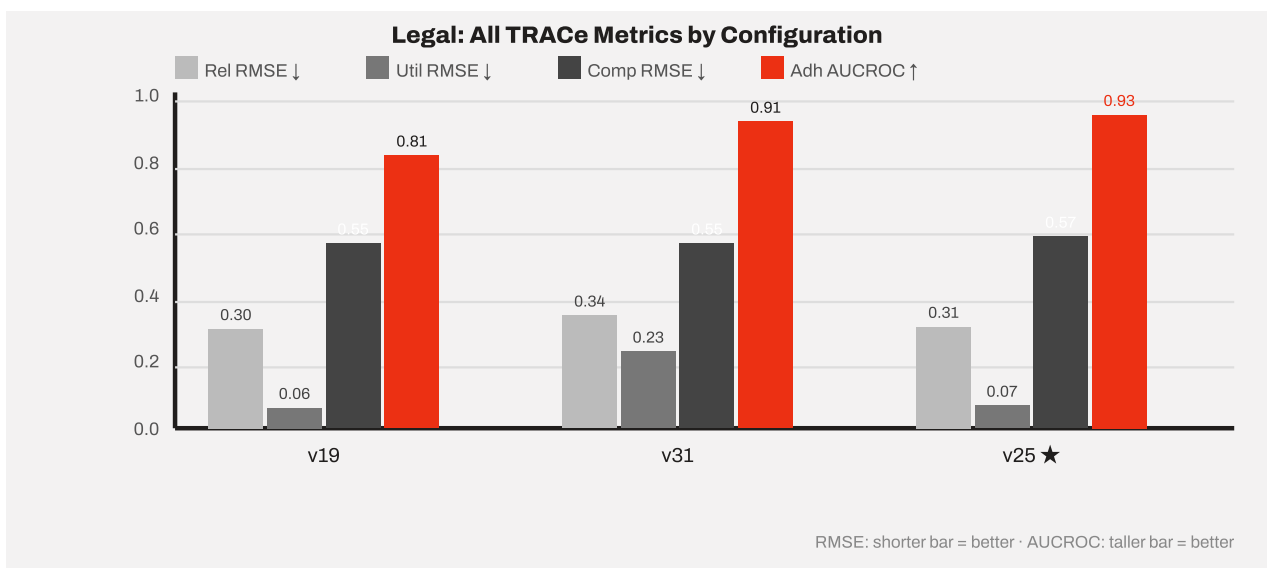


Figure 2. Legal (CUAD): all four TRACe metrics for top-3 configurations. v25 (Long CoT) achieves the highest adherence (0.931); v19 (Rich Dual-Track) achieves the lowest utilization RMSE (0.06). Completeness (~0.55) is structurally high across all configs — a property of dense legal clauses, not a tuning failure.

**v19 (Rich Dual-Track)** achieves near-perfect utilization (0.0599 RMSE) via a JSON-decoupled prompt forcing three components: boolean answer, exact supporting quote, and legal explanation. The Judge evaluates only the extracted content.

**v31 (Paper Short + Parent-Child)** uses 256-token granular indexing for precise semantic matching, while returning full parent documents. High adherence (0.9138) but catastrophic utilization (0.2263) — without structural prompt forcing, high adherence requires verbose over-citing.

**v25 (Long Chain-of-Thought)** achieves the highest adherence (0.9310) by forcing step-by-step reasoning before the answer. CoT functions as a built-in factual verification layer — the model must explicitly cite its evidence source before each claim.

## DOMAIN FINDING — LEGAL

The RAG Tradeoff Triangle governs legal document QA. Standard prompts force a binary choice: high utilization (concise extraction) OR high adherence (verbose, cautious reasoning). Only structural prompt decoupling (Rich Dual-Track) or forced self-verification (Long CoT) can achieve both simultaneously.

### 3.3 Financial Domain (FinQA + TatQA) — Dhananjay 15 ITERATIONS

The financial evaluation ran two tracks: FinQA (6 configs) and TatQA (9 configs). The project's most dramatic single finding emerged from this domain — the catastrophic effect of self-evaluation bias.

#### FinQA — Top 3 Configurations (v3, v4, v5)

| Config     | Key Change                                             | Rel.<br>RMSE | Util.<br>RMSE | Comp.<br>RMSE | Adh.<br>AUCROC           |
|------------|--------------------------------------------------------|--------------|---------------|---------------|--------------------------|
| <b>v3</b>  | Independent judge (gpt-oss-120b) — replaces self-judge | 0.098        | 0.078         | 0.248         | 0.714                    |
| <b>v4</b>  | Independent judge + bge-large-en-v1.5 embedding        | 0.122        | 0.112         | <b>0.186</b>  | <b>0.786</b>             |
| <b>v5*</b> | n=33 diagnostic run                                    | <b>0.089</b> | <b>0.068</b>  | 0.388         | <b>0.496</b> $\triangle$ |

Table 5a. FinQA top-3 configurations. v4 leads on adherence (0.786) and completeness (0.186). v5\* is a diagnostic run (n=33); its adherence of 0.496 falls below random chance — likely due to small sample size, not a genuine capability regression.  $\triangle$  = below 0.5 chance baseline.

#### TatQA — Top 3 Configurations (tatv3, tatv4, tatv10)

| Config            | Key Change                   | Rel.<br>RMSE | Util.<br>RMSE | Comp.<br>RMSE | Adh.<br>AUCROC |
|-------------------|------------------------------|--------------|---------------|---------------|----------------|
| <b>tatv3</b><br>★ | top_k 25/8 + random sampling | 0.157        | 0.160         | 0.397         | 0.558          |
| <b>tatv4</b>      | tatv3 at n=40 (repro. check) | 0.205        | 0.164         | <b>0.389</b>  | <b>0.617</b>   |
| <b>tatv10</b>     | Reverted judge prompt, n=77  | 0.264        | 0.261         | 0.318         | 0.577          |

Table 5b. TatQA top-3 configurations. tatv4 leads on adherence (0.617) and completeness (0.389 RMSE); tatv10 has the lowest completeness RMSE (0.318) with a reverted judge prompt at n=77.

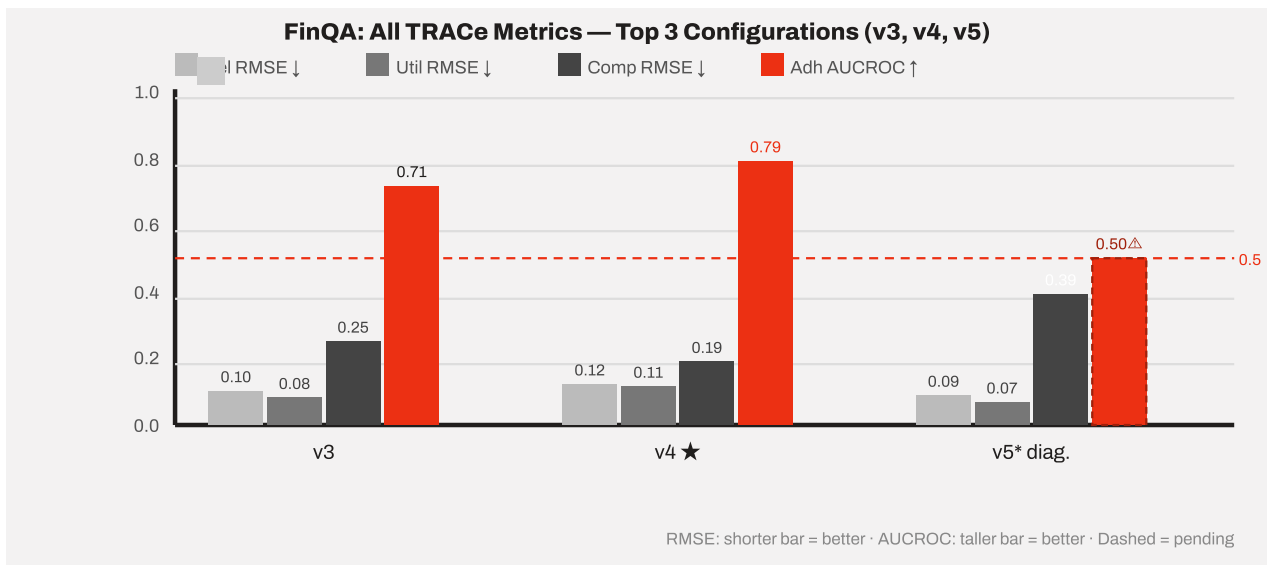


Figure 3a. FinQA: all four TRACe metrics for v3, v4, v5\*. v4 leads on adherence (0.786) and completeness (0.186 RMSE). v5\* is a diagnostic run ( $n=33$ ) — its adherence of 0.496 falls below random chance (dashed border), likely a sample-size artefact rather than a genuine regression.

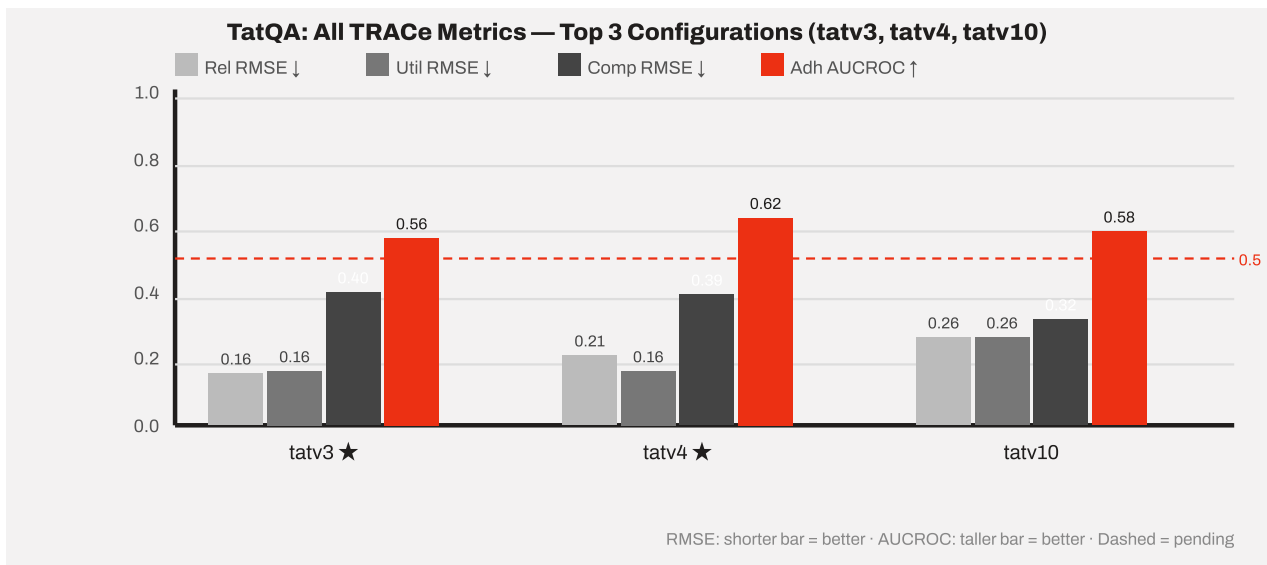


Figure 3b. TatQA: all four TRACe metrics for top-3 configurations. tatv4 leads on adherence (0.617) and completeness RMSE (0.389); tatv10 achieves the lowest completeness RMSE (0.318) with a reverted judge prompt at  $n=77$ .

**v2 (Baseline)** exposed a fundamental flaw: using the same LLM as both generator and judge produces adherence AUCROC of 0.429 — *below random chance*. The model systematically fails to identify its own hallucinations.

**v3** swapped only the judge model (to `openai/gpt-oss-120b`, an independent model family). Results were transformative: Relevance RMSE –67%, Utilization RMSE –78%, Completeness RMSE –44%, Adherence AUCROC +67%. A single architectural decision produced the largest improvement of any single change across all four domains in the entire project.

**v4** added a larger embedding model (`bge-large-en-v1.5`), further improving completeness (–25%) and adherence (+10%), while introducing a genuine tradeoff: relevance and utilization RMSE slightly increased — embedding quality trades precision across metrics rather than uniformly

improving all dimensions.

| Domain | n  | Rejection Rate | Interpretation                                       |
|--------|----|----------------|------------------------------------------------------|
| FinQA  | 15 | 93.3%          | Model correctly refuses 14/15 unanswerable questions |
| TatQA  | 15 | 86.7%          | Model correctly refuses 13/15 unanswerable questions |

Table 6. Negative rejection (RGB-style) for the financial domain. Both exceed 85%, demonstrating graceful degradation when context is entirely irrelevant.

DOMAIN FINDING — FINANCIAL

Self-evaluation bias is catastrophic. Using the same model as generator and judge suppresses hallucination detection to below-random levels (0.429 AUCROC). An independent judge from a different model family is the single most impactful architectural choice for trustworthy RAG evaluation — recovering adherence to 0.714 with no other change.

3.4 General Knowledge Domain (HotpotQA + MSMARCO) — Arhan

41 ITERATIONS

The general knowledge evaluation tested 41 configurations spanning chunking strategies, retrieval architectures, temperature sweeps (0.0–0.8), prompt styles, and sample sizes (n=15–25). This domain uniquely contains both multi-hop reasoning (HotpotQA) and single-hop factoid (MSMARCO) questions, revealing fundamental domain-internal asymmetries.

| Config | Key Change                                                       | Rel. RMSE | Util. RMSE | Comp. RMSE | Adh. AUCROC (pooled) |
|--------|------------------------------------------------------------------|-----------|------------|------------|----------------------|
| v4     | Sentence-window chunk=400, overlap=4, $\lambda$ =0.6, top_k=15/6 | 0.2507    | 0.1910     | 0.4877     | 0.7702               |
| v9     | Pure dense retrieval, no BM25/MMR/reranker                       | 0.2365    | 0.1198     | 0.4202     | 0.6786               |
| v25    | Baseline retrieval, temperature=0.3, n=25, seed=88               | 0.2277    | 0.1252     | 0.3813     | 0.7444               |

Table 7. General Knowledge top-3 configurations with pooled (HotpotQA+MSMARCO) adherence AUCROC.

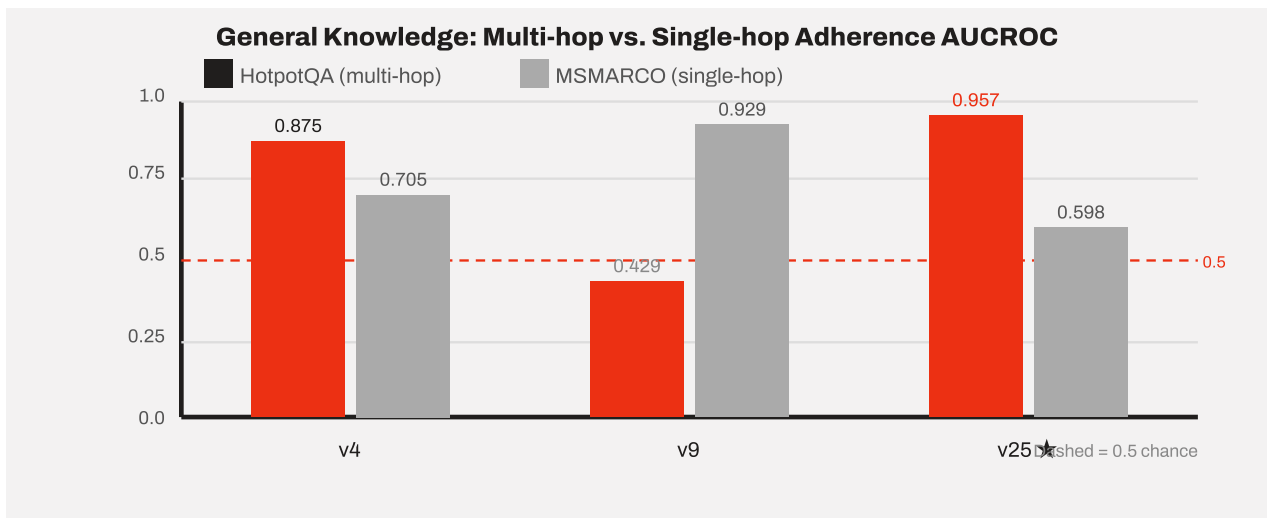


Figure 4a. General Knowledge domain asymmetry: HotpotQA vs. MSMARCO adherence across three configurations. No configuration maximizes both simultaneously.

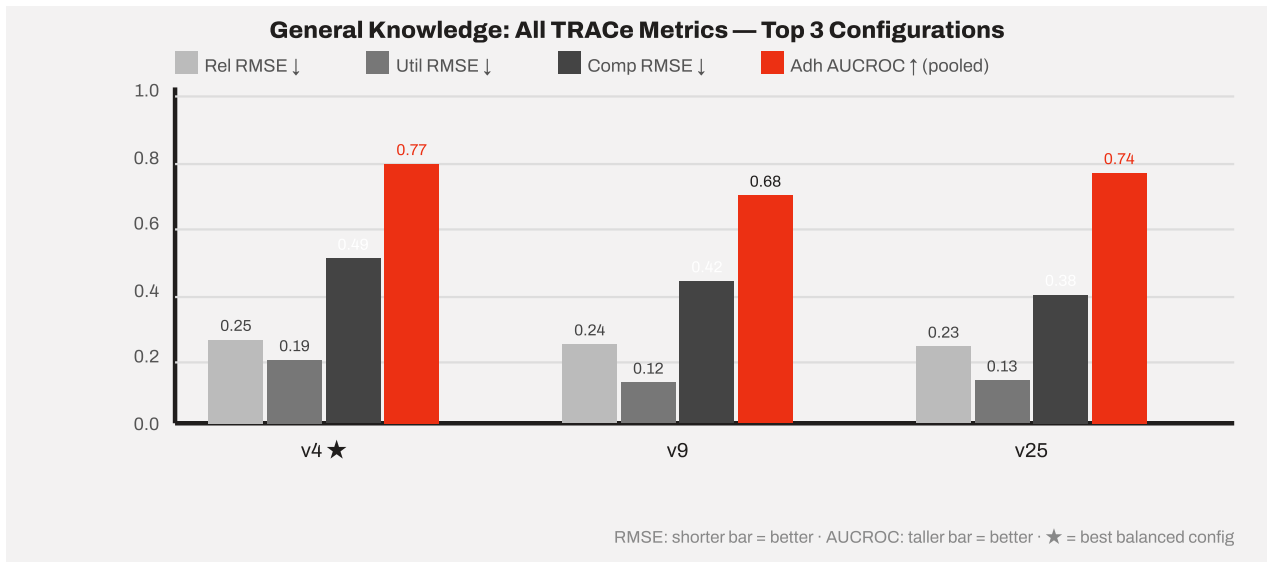


Figure 4b. General Knowledge: all four TRACe metrics for v4, v9, v25. v4 leads on pooled adherence (0.770); v9 and v25 achieve lower RMSE on relevance and utilization but sacrifice either multi-hop or single-hop adherence. v25 achieves the best completeness (0.381 RMSE).

**v4** is the only configuration to exceed 0.7 adherence on both sub-domains simultaneously (HotpotQA: 0.875, MSMARCO: 0.705). Sentence-window chunking with 4-sentence overlap ensures multi-hop evidence passages share boundary context; MMR  $\lambda=0.6$  slightly favours relevance over diversity.

**v9** (pure dense retrieval) reveals the domain asymmetry most starkly: removing BM25 fusion, MMR, and reranking boosts MSMARCO to 0.929 (best for single-hop) while collapsing HotpotQA to 0.429. Multi-hop questions require evidence from semantically different document clusters that pure dense retrieval cannot surface.

**v25** achieves near-perfect HotpotQA adherence (0.957) at  $n=25$ . Temperature=0.3 is the "goldilocks zone" — deterministic enough to stay grounded but flexible enough for natural multi-sentence responses. Temperature=0.0 (fully deterministic) actually performed worse (0.39),

producing mechanically rigid responses the Judge scored inconsistently.

DOMAIN FINDING — GENERAL KNOWLEDGE

Multi-hop and single-hop questions have fundamentally incompatible optimal configurations. v4's balanced profile represents a deliberate architectural compromise, not a globally optimal solution. Seed and sample selection are dominant factors in adherence outcomes at small sample sizes — more dominant than most single-variable configuration changes.

SECTION 4

# Cross-Domain Comparative Analysis

## 4.1 Best Configurations Across Domains

| Domain            | Best Config | Rel. RMSE | Util. RMSE | Comp. RMSE | Adh. AUCROC | Dominant Lever                     |
|-------------------|-------------|-----------|------------|------------|-------------|------------------------------------|
| Biomedical        | v8c         | 0.2756    | 0.1211     | 0.2417     | 0.9231      | MMR redundancy reduction           |
| Legal             | v25         | 0.3055    | 0.0685     | 0.5688     | 0.9310      | Chain-of-thought self-verification |
| Financial         | v4          | 0.122     | 0.112      | 0.186      | 0.786       | Judge independence                 |
| General Knowledge | v25         | 0.2277    | 0.1252     | 0.3813     | 0.744*      | Temperature-controlled generation  |

Table 8. Best configuration per domain with dominant lever. \*Pooled AUCROC; HotpotQA-specific peak 0.957.

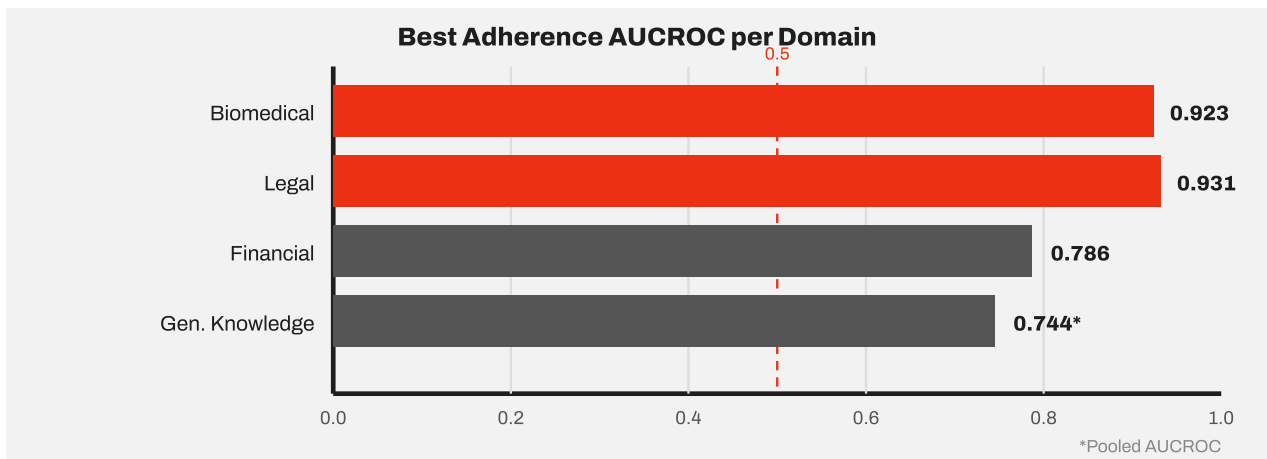


Figure 5. Best adherence AUCROC per domain. Biomedical and Legal both exceed 0.9; Financial and General Knowledge achieve 0.74–0.79. All four substantially exceed the 0.5 random-chance baseline.

## 4.2 Cross-Domain Observations

- **Financial domain achieves the lowest absolute RMSE scores** across all three continuous metrics. This is attributable to the independent, larger judge model providing more precisely calibrated annotations — strongly suggesting that judge quality, not pipeline architecture, is the ceiling-setting variable for evaluation precision.
- **Legal domain exhibits the weakest completeness** (0.5688 RMSE). Legal contracts contain dense, interconnected clauses where the ground-truth "relevant" sentence set is large relative to what any single response can address — a structural property of the domain, not a tunable pipeline deficiency.
- **Biomedical domain achieves the best simultaneous balance** across all four metrics. No other domain has a configuration where all metrics are simultaneously in their competitive ranges.
- **No single configuration generalizes across domains.** Each domain's optimal architecture differs in retrieval strategy, prompt style, temperature, top\_k, and chunking.

### 4.3 Lever Effectiveness Summary

| Lever                 | Biomedical             | Legal                      | Financial                | Gen. Knowledge           |
|-----------------------|------------------------|----------------------------|--------------------------|--------------------------|
| MMR on/off            | ✓ Major (+0.25 AUCROC) | ✓ Helps                    | Not tested independently | Domain-dependent         |
| Independent Judge     | Not tested             | Not tested                 | ✓ Dominant (+67% AUCROC) | Not tested               |
| Prompt style          | "long" > "long_cot"    | Rich Dual-Track / CoT best | Full reasoning required  | CoT stripped performance |
| Temperature           | 0.2 default            | 0.0 default                | 0.0 default              | 0.3 optimal              |
| Chunking              | × Adherence collapse   | Mixed (parent-child helps) | Not needed               | ✓ Helps multi-hop        |
| top_k tuning          | 5 optimal              | 10–15 optimal              | 5 (FinQA) / 8 (TatQA)    | 5–6 optimal              |
| Seed control          | ✓ Essential            | Not tested                 | Confirmed essential      | ✓ Essential              |
| Domain-adapted models | × Regressed            | 70B upgrade helps          | bge-large helps          | N/A                      |

Table 9. Lever effectiveness across domains. ✓ = positive finding; × = negative finding.

### SECTION 5

## Key Findings & Inferences

- 1** **MMR is the single most impactful retrieval lever (Biomedical).** Enabling Maximal Marginal Relevance produced +0.25 AUCROC improvement in adherence — the largest single-lever gain of the entire project. MMR reduces redundancy among retrieved documents, forcing diversity in the generator's context and reducing speculative cross-document synthesis. This improvement came at no cost to any other metric.
- 2** **Self-evaluation bias is catastrophic (Financial).** When the same LLM serves as both generator and judge, hallucination detection falls to 0.429 AUCROC — worse than a coin flip. Swapping to an independent judge recovers adherence to 0.714 with no other change. This is a categorical failure mode rendering self-evaluated RAG systems untrustworthy for any application where factual accuracy matters.



- 3 Reproducibility requires seeded LLM calls (Biomedical, General Knowledge).** Before adding seed=42 to API calls, the biomedical pipeline's apparent best result (v14, adherence 0.9231) was revealed upon seed-controlled retesting (v14b) to be a lucky non-reproducible draw — true seeded performance was 0.5577, a 0.37 AUCROC swing. The entire "chunking fixes adherence" narrative was built on unpinned-seed variance. A noise band of ~0.03–0.08 RMSE and ~0.06 AUCROC was established, below which improvements cannot be distinguished from run-to-run stochasticity.

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- 4 Domain-specific optimal configurations are irreducible (All Domains).** No single configuration achieves competitive performance across all four domains simultaneously. The optimal architectures differ fundamentally in retrieval strategy, prompt style, temperature, top\_k, and chunking — reflecting genuine differences in document length, question complexity, and evidence structure.

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- 5 The Completeness–Adherence tradeoff is structural (All Domains).** Interventions that improve completeness tend to degrade adherence and vice versa. Chunking improves completeness but kills adherence via chunk-boundary recall loss. Widening top\_k improves relevance but introduces hallucination from marginal documents. This tradeoff is an inherent property of the TRACe metric definitions, not a tuning failure.

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- 6 top\_k\_final=5 is a universal sweet spot.** Both narrowing (top\_k=3, insufficient grounding) and widening (top\_k=10, too much marginal content) caused adherence collapse across multiple domains and iterations. Five documents consistently provides the optimal balance between evidence sufficiency and precision.

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- 7 The Judge's scaffolding-sentence leniency is a cross-domain limitation (Financial).** The independent judge inconsistently marks non-factual scaffolding sentences (e.g., "to find X, we need to look at Y") as not fully supported, even while its own category labels imply support. A deterministic rule-based fix was attempted and rejected — it degraded overall AUCROC from 0.786 to 0.321 by also forgiving genuinely hallucinated responses.

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- 8 RAG pipelines fail gracefully under adverse conditions (Financial).** Negative rejection rates of 93.3% (FinQA) and 86.7% (TatQA) demonstrate that the pipeline reliably recognizes when it has no basis to answer and says so, rather than fabricating from parametric memory. Noise robustness sweeps show non-catastrophic degradation — TatQA accuracy holds at 0.80–0.87 across all noise ratios.

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- 9 Benchmark ground truth is not infallible (Financial).** A TatQA ground-truth reference response was found to contain a verifiable arithmetic error — using 2019's supplier percentage where it needed 2018's value. The pipeline's response was verified correct against the source document. At small sample sizes (n=15), even one or two ground-truth errors can meaningfully bias aggregate AUCROC.

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## Conclusions (Part I)

Through 150+ controlled iterations across four specialized domains, several principles emerge as robust, cross-domain findings:

- **Evaluation methodology dominates pipeline architecture.** The financial domain's dramatic improvement from judge independence (+67% AUCROC from a single change) exceeds any retrieval or generation optimization observed in any domain. How you evaluate a RAG system matters more than how you build it.
- **Reproducibility is non-negotiable.** Without seeded LLM calls, experimental conclusions can be entirely built on sampling noise. Any RAG evaluation claiming reproducible results without explicit seed control should be treated with skepticism.
- **Domain expertise informs architecture.** The legal tradeoff triangle, biomedical MMR sensitivity, financial judge-independence requirement, and general knowledge multi-hop/single-hop asymmetry all require domain-specific understanding. Generic "best practices" applied uniformly will underperform.
- **The TRACe framework reveals genuine, irreducible tensions.** Completeness and adherence are in structural opposition; these are fundamental properties of the evaluation geometry, not engineering failures to be fixed.
- **Practical recommendations:** Always use an independent judge; always seed LLM calls; use MMR diversity in retrieval unless domain analysis suggests otherwise; tune top\_k (default 5) per-domain; define metric priority order explicitly for each deployment context.

### SECTION 7

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## References

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10. Live demo for the list of iterations performed - <https://prtvish.github.io/ragbench-top3-demo/rag-demo/>
11. Github link for RAG pipelines (RAGBench & RGB) - <https://github.com/prtvish/ragbench-eval-capstone>

PART II

# RGB Benchmark Evaluation: Model Robustness Across Core RAG Abilities

Evaluating llama-3.3-70b-versatile, llama-3.1-8b-instant, and openai/gpt-oss-20b across Noise Robustness, Information Integration, Counterfactual Robustness, and Negative Rejection

SECTION 8

## Executive Summary

Part II evaluates three contemporary language models against the RGB benchmark framework, testing four core RAG abilities under adversarial conditions. Unlike Part I's pipeline-level TRACe evaluation, RGB measures model-level robustness: how models behave when retrieval is noisy, when documents are factually wrong, or when no useful context exists at all.

TOP-LINE FINDING

All three models substantially outperform the RGB paper's 2023 English baselines (best prior: 31% on Negative Rejection vs. 70–76% here). llama-3.3-70b-versatile **is the strongest overall** — leading on Information Integration and Counterfactual Robustness. A Unicode scoring bug was discovered and corrected for openai/gpt-oss-20b , which had artificially suppressed scores by 50+ percentage points across multiple abilities.

|                                                                                                                                |                                                                                                |                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <p>NOISE ROBUSTNESS</p> <p><b>~97–100%</b></p> <p>All models near-ceiling; ability largely solved at this model generation</p> | <p>INFO INTEGRATION (70B)</p> <p><b>90%</b></p> <p>At noise=0.0; holds to 88% at noise=0.4</p> | <p>NEGATIVE REJECTION</p> <p><b>76.3%</b></p> <p>gpt-oss leads; all beat 2023 baseline of 31%</p> |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|

| Ability                   | Status           | Best model              | Best score |
|---------------------------|------------------|-------------------------|------------|
| Noise Robustness          | PARTIAL          | All comparable          | ≥95.0%     |
| Information Integration   | PILOT (N=50)     | llama-3.3-70b-versatile | 90.0%      |
| Counterfactual Robustness | PILOT (N=50)     | llama-3.3-70b-versatile | 89.7% CR   |
| Negative Rejection        | COMPLETE (N=300) | openai/gpt-oss-20b      | 76.3%      |

Table 10. RGB ability summary. CR = Correction Rate.

SECTION 9

## Project Background & Evaluation Framework

The RGB benchmark defines four core abilities a RAG-capable model must possess to be reliable in production. Unlike TRACe (which measures pipeline quality at the retrieval and generation level), RGB tests model behavior under deliberately adversarial retrieval conditions.

| Ability                   | What it tests                                                                       | Dataset                    |
|---------------------------|-------------------------------------------------------------------------------------|----------------------------|
| Noise Robustness          | Identifies correct answer when irrelevant passages are mixed into retrieved context | en_refine.json             |
| Information Integration   | Synthesizes answers requiring facts from multiple retrieved passages                | en_int.json                |
| Counterfactual Robustness | Detects and corrects deliberate factual errors in retrieved documents               | en_fact.json               |
| Negative Rejection        | Correctly declines to answer when all retrieved passages are irrelevant             | en_refine.json (noise=1.0) |

Table 11. The four RGB abilities and their datasets.

| Model                   | Provider                     | Status    |
|-------------------------|------------------------------|-----------|
| llama-3.3-70b-versatile | Groq (free tier)             | ACTIVE    |
| llama-3.1-8b-instant    | Groq (free tier)             | ACTIVE    |
| openai/gpt-oss-20b      | OpenAI-compatible endpoint   | ACTIVE    |
| gemma2-9b-it            | Groq (deprecated 2025-10-08) | ABANDONED |

Table 12. Model roster. gemma2-9b-it was replaced by openai/gpt-oss-20b after Groq deprecation.

SECTION 10

# Methodology & Experimental Design

| Parameter           | Value                      | Notes                                                   |
|---------------------|----------------------------|---------------------------------------------------------|
| Passages per prompt | 5                          | Fixed across all abilities                              |
| Temperature         | 0.0 (Noise) / 0.2 (others) | 0.0 for fully deterministic noise runs                  |
| Seed                | 2333                       | Consistent across all runs                              |
| Full dataset size   | 300                        | Negative Rejection complete; others at pilot or partial |
| Scoring             | Typography-corrected       | U+202F normalization applied for gpt-oss-20b            |

Table 13. Experimental parameters for the RGB evaluation.

Counterfactual robustness uses four distinct metrics: **Acc** (no documents — tests parametric knowledge), **Acc\_doc** (with counterfactual documents — tests resistance to misleading retrieval), **ED** (error detection rate), and **CR** (correction rate conditional on detection).

## SECTION 11

# Results: Noise Robustness PARTIAL

| Model                   | Noise 0.0 | Noise 0.2 | Noise 0.4 | Noise 0.6 | Noise 0.8 |
|-------------------------|-----------|-----------|-----------|-----------|-----------|
| llama-3.1-8b-instant    | 98.7%     | 98.7%     | 98.7%     | 98.3%     | 95.0%     |
| llama-3.3-70b-versatile | —         | —         | —         | —         | —         |
| openai/gpt-oss-20b      | —         | —         | —         | —         | —         |

Table 14. Noise Robustness accuracy (n=300, llama-3.1-8b-instant only; other runs intentionally skipped due to computational and time constraints).

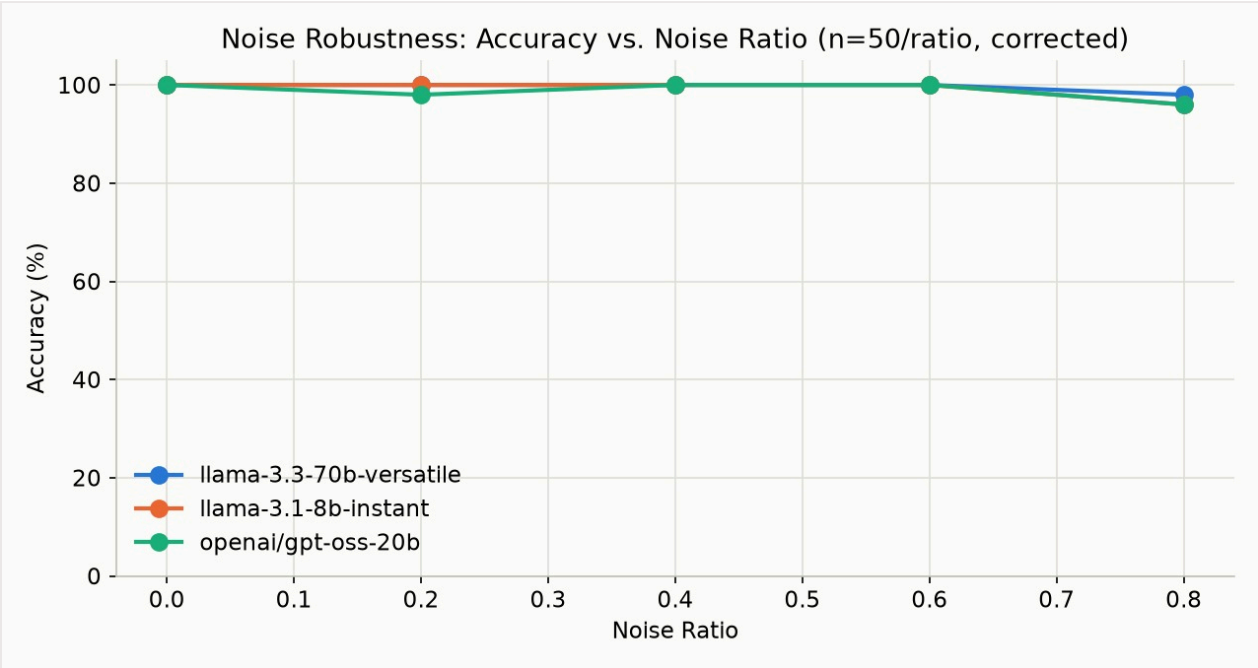


Figure 6. Noise Robustness accuracy across noise ratios (corrected, pilot n=50/ratio). All three models cluster near 95–100%, confirming near-ceiling performance across the noise spectrum.

**CEILING EFFECT**

At n=100, llama-3.1-8b-instant showed near-perfect 100% flat until noise=0.8 (99%). The full n=300 run reveals the true degradation curve: 98.7% through noise=0.4, dropping to 95.0% at noise=0.8. Even with 4 of 5 passages irrelevant, the model finds the correct answer 19 out of 20 times.

SECTION 12

Results: Information Integration PILOT (N=50)

| Model                   | Noise 0.0 | Noise 0.2 | Noise 0.4 | Trend                                               |
|-------------------------|-----------|-----------|-----------|-----------------------------------------------------|
| llama-3.3-70b-versatile | 90.0%     | 84.0%     | 88.0%     | Most resilient; slight dip then recovery            |
| llama-3.1-8b-instant    | 84.0%     | 76.0%     | 62.0%     | Monotonic decline; largest degradation (–22pp)      |
| openai/gpt-oss-20b      | 68.0%     | 78.0%     | 60.0%     | Anomalous rise at 0.2; competitive after correction |

Table 15. Information Integration accuracy (corrected, pilot n=50/ratio). Uncorrected gpt-oss scores were 6–12% — entirely a scoring artifact.

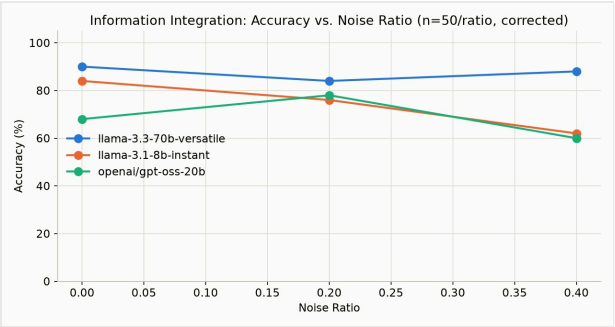


Figure 7a. Corrected scores. gpt-oss is competitive after Unicode normalization.

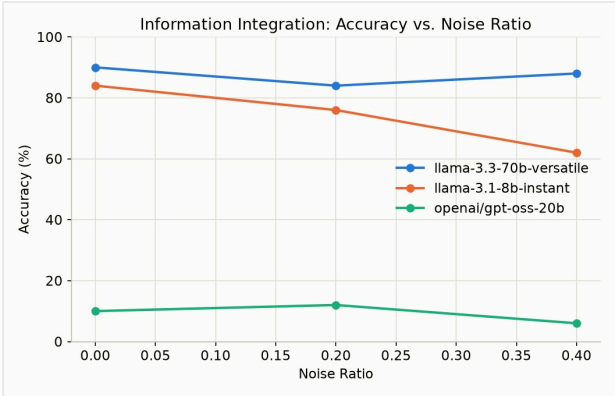


Figure 7b. Uncorrected scores. gpt-oss appeared to score 6–12% — a near-total failure that was entirely a scoring artifact.

SECTION 13

Results: Counterfactual Robustness PILOT (N=50)

| Model                   | Acc (no docs) | Acc_doc (wrong docs) | ED (Error Detect) | CR (Correction) |
|-------------------------|---------------|----------------------|-------------------|-----------------|
| llama-3.3-70b-versatile | 44.0%         | 62.0%                | 58.0%             | 89.7%           |
| llama-3.1-8b-instant    | 64.0%         | 22.0%                | 20.0%             | 80.0%           |
| openai/gpt-oss-20b      | 2.0%          | 18.0%                | 28.0%             | 64.3%           |

Table 16. Counterfactual Robustness (corrected, n=50). gpt-oss Acc=2% is genuine — not a scoring artifact (see Notable Findings below).



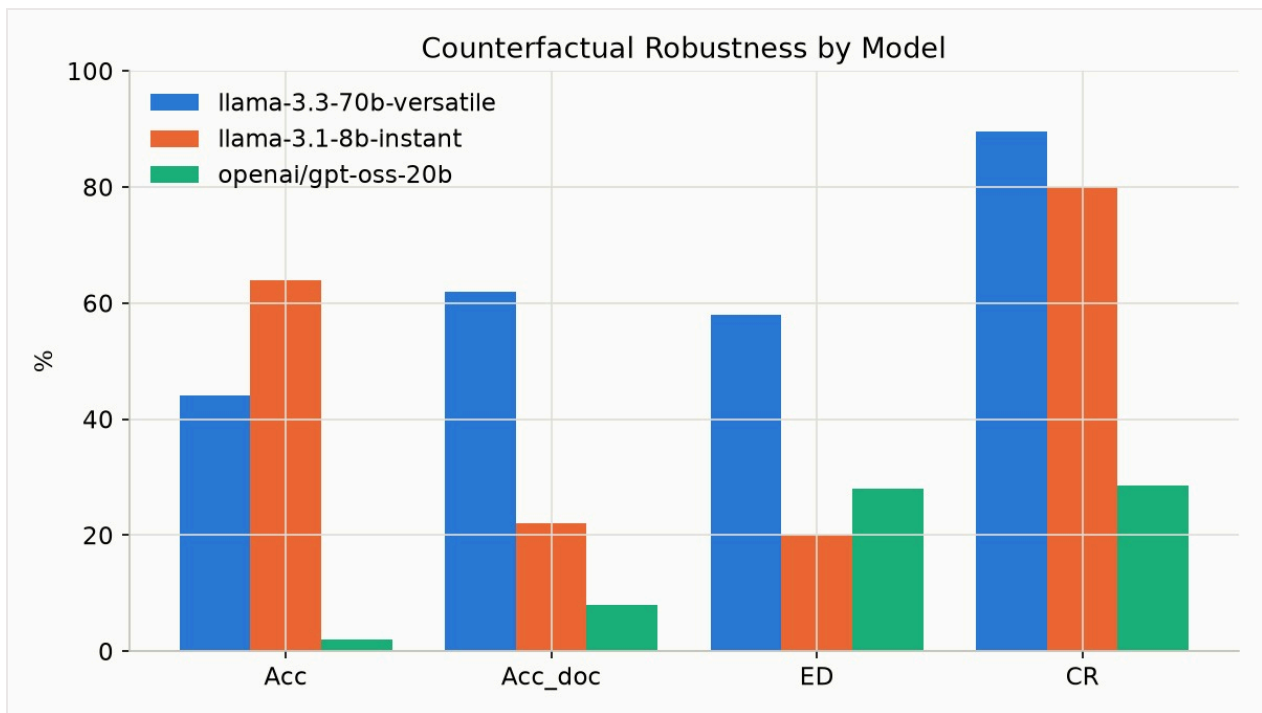


Figure 8. Counterfactual Robustness across all four metrics. The 70B model leads on Acc\_doc, ED, and CR; the 8B model leads on raw parametric accuracy (Acc).

**Notable Finding 1 — Counterfactual priming:** The 70B model scores higher *with* counterfactual documents (62%) than without any documents (44%). A wrong document on a topic appears to prime the model's correct parametric memory about that topic, even while asserting something false. This is a genuine behavioral finding, not an artifact.

**Notable Finding 2 — gpt-oss parametric refusal:** In 49 of 50 zero-document prompts, gpt-oss responds verbatim "I can not answer the question because of the insufficient information in documents." — including on trivial general-knowledge questions ("Super Bowl 2021 location"). This model refuses to answer from parametric knowledge when no documents are provided, regardless of whether the question requires retrieval.

## SECTION 14

## Results: Negative Rejection COMPLETE (N=300)

OPENAI/GPT-OSS-20B

**76.3%**

Best rejection rate; lowest hallucination (7.3%)

LLAMA-3.3-70B-VERSATILE

**74.0%**

Rejection rate

LLAMA-3.1-8B-INstant

**70.0%**

Lowest rejection; highest "other/garbled" (21%)

| Model                   | Rejected (%) | Hallucinated "correct" (%) | Other / garbled (%) |
|-------------------------|--------------|----------------------------|---------------------|
| llama-3.3-70b-versatile | 74.00%       | 10.33%                     | 15.67%              |
| llama-3.1-8b-instant    | 70.00%       | 9.00%                      | 21.00%              |
| openai/gpt-oss-20b      | 76.33%       | 7.33%                      | 16.33%              |

Table 17. Full Negative Rejection breakdown (n=300). "Other/garbled" may include valid refusals phrased differently — true rejection rates are a floor.

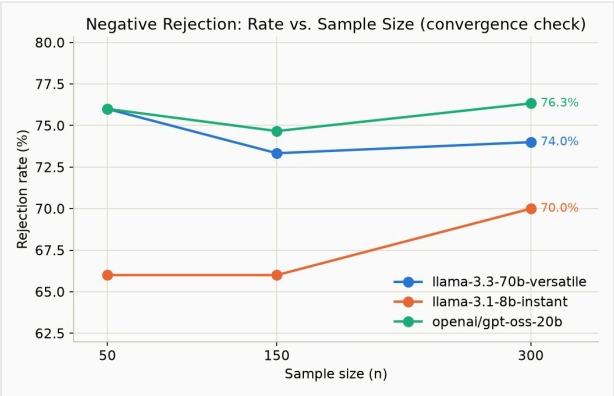
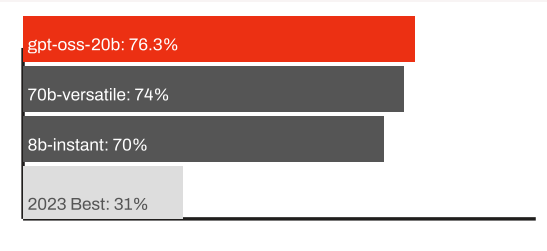


Figure 9. Negative Rejection convergence check: rates wobble at small n (fixed prefix sampling), then stabilize at n=300.

### 2023 Baseline Comparison

The best 2023 RGB paper baseline (English) was 31.00% rejection rate (Qwen-7B-Chat). All three evaluated models exceed 70% — a 2× improvement attributable to newer model generations being better instruction-tuned to refuse unanswerable questions.



### CALIBRATION CAVEAT

gpt-oss-20b leads on rejection rate — but this must be read alongside its Counterfactual Acc of 2%. A model that refuses nearly universally without documents will naturally score well on Negative Rejection. High rejection alone does not prove good calibration; it must be read together with accuracy on abilities where refusing is the wrong answer.

# Cross-Ability Analysis & Model Comparison

| Ability / Metric             | llama-3.3-70b | llama-3.1-8b | gpt-oss-20b | 2023 Baseline |
|------------------------------|---------------|--------------|-------------|---------------|
| Noise Robustness (noise=0.8) | —             | 95.0%        | —           | —             |
| Info Integration (noise=0.0) | 90.0%         | 84.0%        | 68.0%       | —             |
| Info Integration (noise=0.4) | 88.0%         | 62.0%        | 60.0%       | —             |
| Counterfactual CR            | 89.7%         | 80.0%        | 64.3%       | —             |
| Negative Rejection           | 74.0%         | 70.0%        | 76.3%       | 31.0%         |

Table 18. Cross-ability model comparison. Bold = best per metric. 2023 baseline from RGB paper (English).

**llama-3.3-70b-versatile** is the strongest model across the most meaningful abilities. It leads on Information Integration, Counterfactual ED and CR, and is competitive on Negative Rejection. The counterintuitive Acc\_doc > Acc result (62% vs 44%) in Counterfactual is a distinctive behavioral signature worth further investigation.

**llama-3.1-8b-instant** is strong on Noise Robustness and leads on raw parametric accuracy in Counterfactual (64%), but degrades sharply with noise on Information Integration (84% → 62%) and is easily misled by counterfactual documents (Acc\_doc = 22%).

**openai/gpt-oss-20b** is genuinely competitive on Information Integration and Negative Rejection after scoring correction, but the near-universal refusal without documents (Acc=2%) raises calibration concerns that cannot be resolved without further investigation.

SECTION 16

# Engineering Challenges & Lessons Learned

| # | Issue                                     | Impact                                                                                     | Resolution                                                                                                  |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1 | Groq free-tier rate limits (70B model)    | Negative Rejection halted at 42, 176, 216, 221/300 across 5+ attempts                      | Checkpointed harness resumes from last record; no work wasted                                               |
| 2 | gemma2-9b-it Groq deprecation (Oct 2025)  | 400 BadRequestError on every call; 0/300 records possible                                  | Replaced with openai/gpt-oss-20b; model roster rebuilt                                                      |
| 3 | Drive / local storage silent divergence   | Colab Drive and local kernel wrote to different folders; neither knew the other's progress | Audit of both locations; merged more-complete Drive data for 3 abilities; local kept for Negative Rejection |
| 4 | / in model ID breaks checkpoint paths     | gpt-oss-20b checkpoints silently written to wrong locations                                | Added <code>safe_model_name()</code> sanitizer before data loss occurred                                    |
| 5 | Unicode narrow-no-break-space scoring bug | gpt-oss-20b scores collapsed to near-zero across multiple abilities                        | Unicode normalization applied; all affected results recomputed                                              |
| 6 | Small-sample convergence misread          | n=50/100 prefix slices don't converge monotonically                                        | Documented that dataset[:n] is a fixed prefix; only n=300 treated as authoritative                          |

Table 19. Engineering issues encountered during the RGB evaluation phase.

## SHARED LESSON ACROSS BOTH PARTS

Both Part I (seed-control failure, storage divergence) and Part II (Unicode bug, deprecated models) demonstrate the same principle: evaluation harnesses require the same engineering rigor as production systems. The failures were not ML or prompting problems — all were infrastructure plumbing that silently corrupted results until caught by manual inspection.

## SECTION 17

# Scoring Correction: The Unicode Bug

`openai/gpt-oss-20b` outputs used Unicode narrow-no-break spaces (U+202F) and smart punctuation where standard spaces and ASCII punctuation were expected. Exact substring matching failed on these characters, collapsing valid answers to near-zero scores across Information Integration and Counterfactual Robustness.

| Ability / Metric             | Raw (uncorrected) | Corrected    | Change           |
|------------------------------|-------------------|--------------|------------------|
| Info Integration — noise=0.0 | 10.0%             | <b>68.0%</b> | +58pp            |
| Info Integration — noise=0.2 | 12.0%             | <b>78.0%</b> | +66pp            |
| Info Integration — noise=0.4 | 6.0%              | <b>60.0%</b> | +54pp            |
| Counterfactual — Acc_doc     | 8.0%              | <b>18.0%</b> | +10pp            |
| Counterfactual — CR          | 28.6%             | <b>64.3%</b> | +35.7pp          |
| Negative Rejection — Rej     | 76.3%             | 76.3%        | 0pp (unaffected) |

Table 20. Scoring correction impact for openai/gpt-oss-20b. The Negative Rejection headline metric used standard spacing and was unaffected.

## SECTION 18

# Conclusion

Taken together, Part I and Part II of this project paint a consistent and detailed picture of RAG system quality in 2026. The TRACe framework reveals that pipeline-level quality is fundamentally domain-specific and cannot be optimized with a single universal configuration. The RGB framework reveals that modern LLMs are dramatically more robust to adversarial retrieval conditions than their 2023 predecessors, but that robustness varies sharply across model size, ability, and the specific failure mode tested.

Across both frameworks, the most consequential decisions were not ML choices — they were infrastructure and methodology choices. Judge independence, seed control, checkpoint resumability, and careful raw-output inspection each materially changed which conclusions could be trusted. The engineering overhead of getting these right consumed as much time as the experiments themselves, and that time was well spent: without it, several reported numbers would have been wrong by 30–60 percentage points.

The practical recommendation is the same whether building or evaluating a RAG system: treat the evaluation harness as a first-class engineering artifact, invest in reproducibility before optimization, and always read at least a sample of raw outputs before trusting aggregate metrics.

## APPENDIX A

# Documentation Gaps & Open Questions

| # | Gap                                                           | Applies to | Why it matters                                                                                                                         |
|---|---------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Primary capstone PDF unreadable (compressed)                  | Both parts | Background, system design, and implementation sections may contain additional context not captured here                                |
| 2 | Exact prompt templates                                        | Both parts | Results cannot be fully replicated without the verbatim system prompt and user message format                                          |
| 3 | Full Noise Robustness results (70B and Qwen)                  | Part II    | Only 8b-instant has complete n=300 data; 70B and Qwen rows are blank                                                                   |
| 4 | Full-scale Info Integration and Counterfactual                | Part II    | Pilot (n=50) results are directional; guarded by notebook "dont run" placeholders                                                      |
| 5 | Manual review of "other/garbled" Negative Rejection responses | Part II    | 16–21% uncategorized; some may be valid refusals — true rejection rate is a floor                                                      |
| 6 | gpt-oss-20b parametric refusal root cause                     | Part II    | Near-universal refusal without documents may be deliberate alignment or prompt sensitivity — unresolved                                |
| 7 | Part I scoring logic source code                              | Part I     | TRACe metric computation and judge prompt templates not included in the markdown results files                                         |
| 8 | Complete noise robustness data for all Part I domains         | Part I     | RGB-style noise sweep conducted only for Financial domain (FinQA/TatQA); other domains not assessed                                    |
| 9 | PPTX slide content (RAGBench_All_Slides_Final Version 1.pptx) | Both parts | Slide deck was uploaded but could not be parsed; may contain additional diagrams, architecture figures, or narrative not captured here |

Table A1. Documentation gaps. Items 3–5 are resolvable by completing pending experimental runs.